

CrowdStrike Event Log

Add the data into Splunk and set the source type as JSON.

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: crowdstrike_falcon.log

Source type: _json

Save As

	Format	Select...	Select...		
	SourceThreadId	Tags	TargetProcessId	timestamp	TokenType
> Timestamp	22235758	25, 40, 53, 54, 55, 151, 874, 924, 12094627905582, 12094627906234, 263882790666253	4645754484	1658910730575	1
> Advanced	39389444	25, 53, 54, 55, 151, 874, 924, 12094627905582, 12094627906234	4648265060	1658910730575	1

Adversaries usually follow a predictable sequence to steal NTDS.dit.

Reports

Top values

Top values by time

Rare

Events with this field

Values	Count
"cmd.exe" /c "echo Run "Invoke-AtomicTest T1003.003 - TestName 'Create Volume Shadow Copy with vssadmin'" to fulfill this requirement"	1
"cmd.exe" /c "vssadmin.exe create shadow /for=C: & mklink /D C:\Temp\vsstore \?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1"	1
"cmd.exe" /c "vssadmin.exe create shadow /for=C:"	1
vssadmin.exe create shadow /for=C:	1
vssadmin.exe create shadow /for=C:	1

As a result, we extracted the CommandLine field and identified several interesting commands.

Command

Analysis

/c "echo Run "Invoke-AtomicTest T1003.003 -TestName 'Create Volume Shadow Copy with vssadmin'" to fulfill this requirement"

breadcrumb left by the Atomic Red Team test framework. It is not malicious by itself, but it identifies the attack simulation that follows.

/c "vssadmin.exe create shadow /for=C: & mklink /D C:\Temp\vssstore \?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1"

core activity of the attack chain.

/c "vssadmin.exe create shadow /for=C:"

py creation command.

xe create shadow /for=C:

py creation command.

The command below represents the main attack activity:

```
cmd.exe /c "vssadmin.exe create shadow /for=C: & mklink /D C:\Temp\vssstore \?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1"
```

This command performs two actions:

1. vssadmin creates a Volume Shadow Copy of the C: drive where NTDS.dit is stored.
2. mklink creates a symbolic link to the shadow copy, making it accessible through C:\Temp\vssstore.

This technique allows an attacker to access files such as NTDS.dit even when they are locked by the operating system.

The standalone command:

```
cmd.exe /c "vssadmin.exe create shadow /for=C:"
```

appears to be an earlier execution or test before the combined command was executed. The timestamps show these events occurred shortly before the command that created the symbolic link.

Another important finding was the ProcessRollup2 event. This event is similar to Windows Event ID 4688 and records process creation activity.

```
    "id": "9b9afac9-0d86-11ed-b9af-0248097df5d7",
    "name": "ProcessRollup2V19",
    "timestamp": 1658910730575,
    "event_simpleName": "ProcessRollup2",
    "event_platform": "Win",
    "cid": "124cb22314bf4f519be84bce582e7a6b",
    "ip": "5.75.7.207"
  }
```

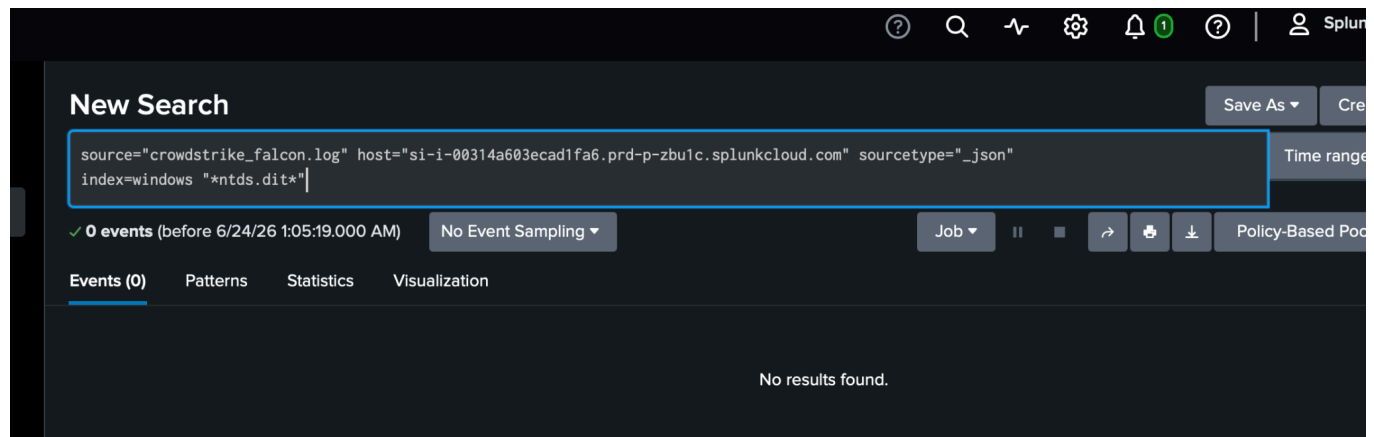
Using ProcessRollup2 events, we confirmed that the same user account (AuthenticationId: 2427434) executed:

```
cmd.exe /c "vssadmin.exe create shadow /for=C: & mklink /D C:\Temp\vssstore  
\?\GLOBALROOT\Device\HarddiskVolumeShadowCopy1"
```

This provides enough evidence to escalate the alert to the Incident Response team for further investigation.

To verify whether the NTDS.dit file was accessed, I searched for references to the file in the CrowdStrike logs.

```
source="crowdstrike_falcon.log" host="si-i-00314a603ecad1fa6.prd-p-zbu1c.splunkcloud.com"  
sourcetype="_json"  
index=windows "ntds.dit"
```



No events containing NTDS.dit were identified. The available logs do not provide enough visibility to confirm direct access to the file.

Conclusion

Based on the CrowdStrike logs, the evidence suggests activity related to MITRE ATT&CK T1003.003 (NTDS Credential Dumping). The logs showed the use of vssadmin.exe to create a shadow copy and mklink to make the shadow copy accessible. These commands are commonly used to access NTDS.dit. While the logs did not show direct access to the NTDS.dit file, the process activity provides enough evidence to escalate the event for further investigation.